3. Goodness of fit tests

These are used to test if a sample can be reasonably regarded as coming from a specified distribution.

χ^2 goodness of fit tests

Hypotheses:

Case (i):

The parameters of the distribution are fully specified in the null hypothesis.

H_0 : The data come from a distribution with probability function $f(x; \theta_0)$, where θ_0 is a specified (vector of) parameter(s).

H_1 : The data don't follow this distribution.

Case (ii):

The distribution is not fully specified.

H_0 : The data come from a distribution with probability function $f(x; \Theta)$

H_1 : The data don't follow this distribution.

The test is based on the χ^2 statistic given as

$$X^2 = \sum_{i=1}^{k} \frac{(f_i - e_i)^2}{e_i}$$

where f_i and e_i are observed and expected frequencies as determined below.

Discrete distributions

The random variable under consideration takes k discrete values, with observed frequencies f_i and expected frequencies (under the model specified in H_0) e_i, $i = 1, \ldots, k$.

Continuous distributions

Divide the range of the random variable into k cells; f_i are the observed data frequencies in cells $i = 1, \ldots, k$, and e_i are the corresponding expected frequencies under the hypothesised distribution (obtained using the cumulative distribution function).

Discrete distribution:

Accidents	Observed
0	7
1	8
2	13
3	10
>=4	12

Continuous distribution:

Age	Frequency
0 - 9	7
10 - 19	5
20 - 29	4
30 - 39	6
40 - 49	1
50 - 59	2

In both cases, we have that under H_0 (approximately)

$$X^2 \sim \chi^2_{k-d-1}$$

where d is the number of model parameters estimated using the data (depending on H_0 in case (i) or (ii) above).

This sampling distribution can be used to obtain p-values (or critical values) for testing.

Reject H_0 if the differences between observed and expected frequencies are large, i.e. if the value of the χ^2 statistic is large with reference to its sampling distribution.

Note: The value of X^2 is sensitive to small values of e_i, in which case the approximate χ^2_{k-d-1} sampling distribution is less valid. In practice, we combine adjacent cells so that $e_i \geq 5$ for all i.

Example 2.1 (cont.):

Test the hypothesis that the Stamford ozone data come from a normal distribution.

Solution: ...

Test also whether a normal distribution fits adequately a suitable transformation of the data. [*Exercise*]

Example 2.1 (cont.) χ^2-test

Cell	f_i	e_i	$(f_i - e_i)^2/e_i$
[0,25)	7	8.531	...
[25,50)	26	15.246	
[50,75)	29	21.785	
[75,100)	25	24.886	
[100,125)	17	22.729	
[125,150)	9	16.597	
[150,175)	8	9.689	
[175,200)	3	4.522	6.712
[200,225)	6	1.687	
[225,250)	2	0.503	

$\chi^2 = 31.458$

With last 3 cells grouped: $\chi^2 = 18.205$

Before adjustment

Cell	fi	ei	(fi-ei)^2/ei
[0, 25)	7	8.531	0.274758059
[25, 50)	26	15.246	7.585498885
[50, 75)	29	21.785	2.389544411
[75, 100)	25	24.729	0.002969833
[100, 125)	17	22.729	1.444033657
[125, 150)	9	16.597	3.477400072
[150, 175)	8	9.689	0.294428837
[175, 200)	3	4.522	0.512269792
[200, 225)	6	1.687	11.02665619
[225, 250)	2	0.503	4.455286282
		sum	**31.46284602**

$$\chi^2 = \sum_{i=1}^{k} \frac{(f_i - e_i)^2}{e_i} = 31.4628 \sim \chi^2_{10-2-1=7}$$

$$\text{p-value} = P(\chi^2_7 > 31.4628) = 5.1 \times 10^{-5}$$

After adjustment

Cell	fi	ei	(fi-ei)^2/ei
[0, 25)	7	8.531	0.274758059
[25, 50)	26	15.246	7.585498885
[50, 75)	29	21.785	2.389544411
[75, 100)	25	24.729	0.002969833
[100, 125)	17	22.729	1.444033657
[125, 150)	9	16.597	3.477400072
[150, 175)	8	9.689	0.294428837
[175, 250)	11	6.712	2.739413588
		sum	**18.20804734**

$$\chi^2 = \sum_{i=1}^{k} \frac{(f_i - e_i)^2}{e_i} = 18.208 \sim \chi^2_{8-2-1=5}$$

$$\text{p-value} = P(\chi^2_5 > 18.208) = 0.0027$$

In R

```
ozone = dget("ozone.R")
attach(ozone)
stamford
xbar= mean(stamford); xbar
sdbar= sd(stamford); sdbar

### Use cells of length 25
stam.breaks = seq(0,250,by=25); stam.breaks                ## define breaks
stam.cut = cut(stamford, breaks = stam.breaks, right=F);  ## bin data
stam.table = table(stam.cut); stam.table                   ## count data in bins

### Compute expected frequencies
prob = numeric(10);  exp.f = numeric(10)
for (i in 1:(length(stam.breaks)-1)) {
  prob[i] = pnorm(stam.breaks[i+1], xbar, sdbar) - pnorm(stam.breaks[i], xbar, sdbar)
  exp.f[i] <- prob[i] * length(stamford)
}
exp.f

### Extract observed frequencies
obs.f = numeric(10)
for (i in 1:10) obs.f[i] = stam.table[[i]]
obs.f
### or
obs.f = hist(stamford, breaks=stam.breaks, plot=F, right=F)$counts
```

```
### Compute chi-sq statistic
x2 = sum( (obs.f - exp.f)^2 / exp.f)
x2

### Calculate p-value;
pval = 1 - pchisq(x2, (10-2-1))
pval

##############################################
### Now join cells together
exp.f1 = c(exp.f[1:7], sum(exp.f[8:10])); exp.f1
obs.f1 = c(obs.f[1:7], sum(obs.f[8:10])); obs.f1

### Perform test again
x2.new = sum( (obs.f1 - exp.f1)^2 / exp.f1); x2.new

pval.1 = 1 - pchisq(x2.new, (8-2-1))
pval.1
```